

Setup Guide: Obstacle Avoidance

Time Allocation: 90 Minutes

Challenge Overview

The robot must detect and drive around 3 randomly placed cardboard obstacles and reach the finish zone.

Environment & Arena Setup

- **Arena Dimensions:** Clear a flat, smooth floor space measuring approximately 3 meters by 3 meters to serve as the testing arena.
- **Start and Finish Zones:** Use colored tape (e.g., green tape) to mark a 50cm x 50cm **Start Zone** in one corner of the arena. Mark a corresponding **Finish Zone** in the opposite corner using red tape.
- **Obstacle Placement:** Gather 3 standard cardboard boxes (roughly 15cm x 15cm x 15cm). Randomly place these three obstacles within the central navigation path between the start and finish zones, ensuring there is at least enough clearance for the robot to navigate between them.
- **Safety and Boundaries:** Use masking tape or walls to outline the outer perimeter of the arena so robots do not wander off into other classroom spaces.